

Certified Potts Filters and Order-Controlled Algebra for Unknot Recognition

**Exact arithmetic, component factorization, adversarial examples,
and a precise research program toward quasi-polynomial complexity**

Research continuation for the ProveIt project

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Abstract

We develop and implement a certified scalar preprocessing improvement for ProveIt’s unknot recognizer. A signed checkerboard Potts transfer replaces the current Jones filter’s arbitrary matching frontier by equality patterns of a fixed number of spin colors. For every crossing order of a connected plane knot diagram, the number of active vertices of either Tait graph is at most half the number of cut diagram edges. This yields $\text{poly}(n, w)q^{w/2}$ arithmetic work at fixed integer q , with all normalization and coefficient lengths accounted for. Independent processed components can be represented by separate tensors; a relative-color alignment formula then gives a stronger bound in terms of the largest component frontier.

For $q \geq 5$, computations take place exactly in $\mathbb{Z}[x]/(x^2 - (q - 2)x + 1)$, using integer pairs and no floating point. An essential adversarial calculation shows that the previously studied $q = 5$ specialization is identically equal to the unknot value on an infinite family of nontrivial weaving knots. Every exact $q \geq 6$ specialization detects every nontrivial knot in that family. This motivates $q = 6$ for the new exact backend without asserting that a single Jones value detects all knots.

We also prove a residue characteristic-polynomial annihilator and sharp radical-matrix nilpotence bounds over truncated polynomial rings. These give short inverse and Fitting certificates for supplied same-matching endomorphisms, extending the scope of earlier scalar splitting. The package includes implementations, integration changes, independent smoothing-cube and algebraic oracles, raw benchmark records, and reproducible certificate replay. The Potts correspondence and single-exponential width algorithms have substantial prior literature. Our contribution is their precise, audited implementation and the associated scope analysis for this repository. A complete quasi-polynomial unknot recognition bound is not established.

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1 Outcome, scope, and the audited baseline

The immediate outcome is a new family of exact Jones obstruction backends, together with a component factorization that improves their dependence on the geometry of a supplied crossing order. These are useful pieces of a complete recognizer. They are not, by themselves, complete recognition algorithms: inequality with the unknot’s Jones value proves knottedness, whereas equality requires another method.

This distinction determines both the software interface and the mathematical claims. In the implementation, a successful obstruction returns `KNOTTED`. Equality returns an inconclusive result. Exhausting a local filter budget also returns an inconclusive result, with its reason recorded; it does not produce a topological verdict. The enclosing recognizer retains its existing complete Khovanov fallback, subject to the user’s global resource limits. Thus a bounded execution can still return `UNKNOWN`.

1.1 What was inspected and preserved

The baseline is ProveIt commit `86f23490006c880cd01db9bec10761d39c6395c8`, dated 8 October 2026 [21]. The source snapshot includes the Python `fastunknot` package, its fixtures and tests, the synthesis sources, and the report summaries. Downloaded source blobs were checked against their Git object identifiers. The included reference implementation is immutable within the experiment package. The baseline test suite passed all 134 tests.

The baseline already contains substantial work: short-braid certificates, Seifert and descending-diagram tests, Reidemeister simplification, connected-sum handling, modular Alexander and Jones obstructions, optional exact Alexander computation, and several exact Khovanov scanner representations. Earlier research also introduced canonical multiplicities, shared morphisms, radical perturbation inverses, saturated or Euler-assisted terminal decisions, homological windows, and scalar Fitting decompositions [15–20]. Those ideas remain predecessors, rather than contributions claimed here. The immediately preceding continuation’s scalar splitting and common-nilpotent decision shortening were inspected separately; the ring-valued extension in Section 9 is carefully distinguished from them.

The current Jones filter stores boundary smoothing matchings and their scalar coefficients. For an arbitrary cut with w endpoints, a universal matching count is $(w - 1)!!$, not a Catalan number. A Catalan bound requires additional boundary-order geometry that an arbitrary crossing prefix need not possess. This creates an opportunity to change the scalar representation without making an invalid noncrossing assumption.

1.2 Claims and their evidential status

Claim	Status and limit
Signed Tait/Potts expression for Jones values	Classical; derived in the repository’s normalization and checked against an independent smoothing cube.
At most $w/2$ active Tait vertices	Proved here by a direct boundary-transition count for every crossing order of a connected plane knot shadow.
Fixed- q equality-pattern transfer	Implemented and verified; costs $\text{poly}(n, w)q^{w/2}$, with linear-in- n coefficient bit lengths for fixed q .
Independent-component transfer	Implemented; a relative-color merge restores precisely the information needed when components first interact.

Claim	Status and limit
$q = 5$ blind weaving family	Proved by specializing the classical three-braid trace formula; preserved as an adversarial regression family.
$q \geq 6$ detects that family	Proved; says nothing universal about other knot families.
Residue-polynomial Fitting certificates	Proved for matrices over specified commutative truncated rings; verified by a separate algebraic oracle.
Quasi-polynomial full recognition	Conditional research goal. No complete complexity proof or complete quasi-polynomial implementation is supplied.

Table 1: The claim ledger. A proof supplied in this article is not a claim of literature-wide priority.

1.3 Prior literature and the nature of the improvement

The Potts formulation of the Jones polynomial belongs to the classical state-model literature [4, 13, 23]. Meichanetzidis and Kourtis explicitly used the same integer-color specializations, including $q = 5$, in tensor-network evaluation [12]. Makowsky established bounded-treewidth tractability for signed Tutte and Kauffman computations [10]. Maria gave single-exponential width dependence for general quantum link invariants [11]. The broad statement that Jones evaluation is single exponential in a suitable width is therefore not a new result.

The contribution appropriate to this continuation is more concrete. We derive a bound directly in the crossing-order parameter used by ProveIt, make equality-pattern aggregation and normalization explicit, implement exact quadratic arithmetic, retain resource-safe recognition semantics, identify a structural failure of the $q = 5$ point, and supply measurements including regressions. The component factorization and residue-algebra sections give additional, independently checkable improvements whose exact applicability is stated.

The public `openai/math` collection was inspected as suggested by the user [14]. It did not supply a missing unknot-recognition theorem. Its useful role here is methodological: formulate explicit lemmas, test a proposed general claim on adversarial families, and preserve enough artifacts that independent readers can reproduce the calculation. None of the results in this article is attributed to that collection without a specific mathematical dependency.

1.4 Complexity terminology

Throughout, n is the number of crossings in the supplied diagram, w is a maximum cut-edge count for a crossing order, and q is an integer color count. A statement for fixed q does not hide a growing input parameter. We use *quasi-polynomial* in the broad sense $\exp((\log(n+2))^{O(1)})$, and state the stronger bound $n^{O(\log n)}$ explicitly when it is obtained. Arithmetic-operation bounds and bit-operation bounds are separated. Timing improvements on a finite corpus are not asymptotic evidence by themselves.

2 The signed Tait partition function in the repository convention

2.1 Input, embedding, and signs

Let D be a classical one-component knot diagram with $n > 0$ crossings. Its shadow is a connected plane four-valent multigraph. The public input validator checks that the supplied planar-diagram

code has the required edge incidences, one link component, and spherical rotation system. The following construction uses that embedding, not an abstract graph with its rotation data discarded.

Choose one checkerboard color. The Tait graph $G = (V, E)$ has one vertex for each region of that color and one edge for each crossing. It can contain loops and parallel edges. Write $v = |V|$. At a crossing e , designate as the *omitted* smoothing the one that leaves the two selected-color corners disconnected across that crossing. Its bracket coefficient is A^{ϵ_e} , where $\epsilon_e \in \{1, -1\}$. The included smoothing has coefficient $A^{-\epsilon_e}$. This convention, which is checked from the local darts in the implementation, avoids assuming that an alternating-diagram sign rule applies to all diagrams. Put

$$s = \sum_{e \in E} \epsilon_e, \quad \delta = -A^2 - A^{-2}.$$

We use the *full* bracket $B_D(A)$, for which an isolated circle has value δ . The normalized Jones polynomial is

$$V_D(A^{-4}) = \frac{B_D(A)}{\delta(-A^3)^{\text{wr}(D)}}. \quad (1)$$

Consequently the full bracket of an unknot diagram with writhe $\text{wr}(D)$ is $\delta(-A^3)^{\text{wr}(D)}$. Losing the initial circle factor would invalidate the obstruction.

Lemma 2.1 (State circles). *For $S \subseteq E$, let $k(S)$ be the number of components of the spanning subgraph (V, S) , including isolated vertices. The associated smoothing state has $c(S) = 2k(S) - v + |S|$ circles.*

Proof. A regular neighborhood of (V, S) in the sphere retracts to that subgraph, so its Euler characteristic is $v - |S|$. Its $k(S)$ components are planar surfaces. If their total number of boundary components is c , their Euler characteristic is $2k(S) - c$. Those boundary components are the smoothing circles. Equating the two expressions proves the formula. An isolated vertex has a disk neighborhood and contributes one circle, as required. Loops and parallel edges are included in the same argument. $\square$

2.2 From smoothing states to spins

The multivariate random-cluster partition function is

$$Z_G(Q, \{z_e\}) = \sum_{S \subseteq E} Q^{k(S)} \prod_{e \in S} z_e.$$

At a positive integer $Q = q$, the Fortuin–Kasteleyn identity is

$$Z_G(q, \{z_e\}) = \sum_{\sigma: V \rightarrow [q]} \prod_{e=uv} (1 + z_e[\sigma(u) = \sigma(v)]). \quad (2)$$

To verify it, expand the product on the right. For each selected subset S , the spin assignment must be constant on each component of (V, S) , giving exactly $q^{k(S)}$ choices. This also proves the identity for loops, parallel edges and isolated vertices [23].

Proposition 2.2 (Signed partition-function formula). *In the rational-function field, or after localizing at δ , with $Q = \delta^2$ and $z_e = \delta A^{-2\epsilon_e}$,*

$$B_D(A) = \delta^{-v} A^s Z_G(Q, \{z_e\}). \quad (3)$$

Proof. The smoothing indexed by S contributes

$$A^s \prod_{e \in S} A^{-2\epsilon_e} \delta^{2k(S)-v+|S|}.$$

Factor out $\delta^{-v} A^s$ and sum. The remaining term for S is $(\delta^2)^{k(S)} \prod_{e \in S} (\delta A^{-2\epsilon_e})$. $\square$

Set $x = A^4$. Then

$$Q = x + 2 + x^{-1}, \quad z_e = -1 - x^{-\epsilon_e}.$$

Thus the spin interaction has the especially simple form

$$h_e(a, b) = \begin{cases} -x^{-\epsilon_e}, & a = b, \\ 1, & a \neq b. \end{cases} \quad (4)$$

This is the signed interaction used in the earlier tensor-network formulation [12]. It allows all crossings to be processed as local equality tests, without tracking smoothing connectivity.

2.3 Eliminating fourth roots from normalization

Since $\delta = -(x+1)A^{-2}$, substituting (3) into (1) gives

$$V_D(A^{-4}) = (-1)^{\text{wr}(D)+v+1} A^E \frac{Z_G}{(x+1)^{v+1}}, \quad E = s - 3 \text{wr}(D) + 2(v+1). \quad (5)$$

Lemma 2.3 (Integral normalization exponent). *For a knot, E is divisible by four.*

Proof. For a smoothing state with coefficient exponent s_σ and circle count c_σ , the residue

$$s_\sigma - 3 \text{wr}(D) + 2(c_\sigma - 1) \pmod{4}$$

is unchanged when one smoothing is flipped: the exponent changes by ± 2 and the circle count by ± 1 . In the oriented Seifert state, $s_\sigma = \text{wr}(D)$. The genus formula for a one-component Seifert surface gives $c_\sigma - 1 \equiv n \equiv \text{wr}(D) \pmod{2}$, so the residue is zero. In the all-omitted state, $s_\sigma = s$ and $c_\sigma = v$. Its expression differs from E by four. $\square$

Writing $h = E/4$, the exact equality test can therefore be performed as

$$Z_G \stackrel{?}{=} Z_U, \quad Z_U = (-1)^{\text{wr}(D)+v+1} x^{-h} (x+1)^{v+1}. \quad (6)$$

No fourth root of x is needed. This is an equality of algebraic integers in the ring introduced in Section 4. The comparison is obtained by multiplying through the nonzero normalization factor; the implementation never needs to divide by $x+1$.

The empty crossing diagram is handled separately as a single circle, with $Z_G = Z_U = q$. For a link with ℓ components the parity residue in the lemma would instead be $2(\ell-1) \pmod{4}$. This article and the public recognizer require a knot; accepting all links without revisiting normalization would be an unsupported generalization.

3 Arbitrary crossing orders and equality-pattern states

3.1 A local half-frontier theorem

Fix a permutation of the n crossings, and let P_i be its first i crossings. Define w_i to be the number of diagram edges with one endpoint in P_i and the other outside P_i . A loop at one crossing is never counted. The scan width is $w = \max_i w_i$.

In the selected Tait graph, let f_i be the number of vertices incident with both a processed and an unprocessed Tait edge. These are precisely the spin variables that must survive after step i . A vertex is introduced at its first incident edge and forgotten after its last.

Theorem 3.1 (Half-frontier bound). *For every crossing order of a connected plane knot shadow and either fixed checkerboard color, $2f_i \leq w_i$ for every i .*

Proof. The boundary of each face of a connected plane graph is a single cyclic dart walk. This remains true when a boundary repeats vertices or edges. On the walk around a selected-color region, mark each crossing incidence according as that crossing belongs to P_i or its complement. An active Tait vertex corresponds to a walk containing both marks. A cyclic binary word with both symbols has at least two transitions.

Each transition traverses a shadow edge with endpoints of opposite status. Conversely each such cut edge occurs on exactly one selected-color face boundary. Count the transitions over all active selected-color regions. There are at least $2f_i$ of them and at most w_i . This proves the claim. $\square$

The result does not assume that the processed diagram is a disk tangle. Repeated face incidences, Tait loops, parallel edges and articulation points do not invalidate the proof. Connectedness matters: a face of a disconnected plane graph can have several boundary walks, each monochromatic in status, while their union contains both statuses. The one-knot input condition avoids that case. Classical relations between a plane graph and its medial graph provide broader width context [22]; the direct proof above supplies the exact inequality used by this implementation.

Both checkerboard colors satisfy the theorem. The code computes their profiles and chooses a single color minimizing the pair $(\max_i f_i, \sum_i f_i)$, with a deterministic tie break. It does not switch colors at individual steps. A pointwise minimum of two profiles would require a tensor conversion that has not been constructed.

3.2 Quotienting names of colors

The Potts interaction depends only on equality of spins. The symmetric group S_q acts on every spin assignment by relabeling colors and preserves every weight. On an ordered active set of f vertices, an orbit is a set partition into at most q blocks. Encode it by its restricted-growth word, for example 0102 for a pattern whose first and third vertices have equal spins and whose other two spins are distinct from those and from each other.

Let

$$B_q(f) = \sum_{k=0}^{\min(q,f)} \left\{ \begin{matrix} f \\ k \end{matrix} \right\} \leq q^f, \quad B_q(0) = 1. \quad (7)$$

For fixed q and large f , $B_q(f) \sim q^f / q!$. These are partitions by *color equality*, not partitions by graph connectivity and not planar noncrossing partitions.

Definition 3.2 (Aggregate coefficient). The coefficient of an equality pattern is the sum of the partial Potts weights over all labeled spin assignments on the introduced vertices whose restriction to the active set has that pattern. Spins on forgotten vertices have already been summed out.

The word *sum* is essential. A per-representative value and an orbit-aggregate value obey different introduction rules.

Proposition 3.3 (Exact transfer rules). *The following operations maintain the aggregate-coefficient invariant over any commutative coefficient ring with a specified unit x .*

1. *To introduce a vertex when k color classes are active, create one extension for each existing class with multiplier 1, and, if $k < q$, one new-class extension with multiplier $q - k$.*
2. *To process an edge, multiply by $-x^{-\epsilon_e}$ if its endpoint classes agree and by 1 otherwise.*
3. *To forget vertices, delete their positions, canonically relabel the remaining word, and add coefficients with the same resulting word.*

Proof. For each labeled active assignment, a new spin has exactly one choice for each currently used color and $q - k$ choices among unused colors. This proves the first rule. The edge weight is constant on an equality orbit, proving the second. The last rule is the pushforward of a finite sum under restriction to a smaller active set. Once a vertex is forgotten, no future edge is incident with it, so its color name imposes no further condition. $\square$

At a crossing at most two Tait vertices are introduced. The implementation combines introduction, multiplication and forgetting, inserting results directly into the following table. It records transient insertions as well as completed tables when reporting its maximum state count. A scalar coefficient that is exactly zero is removed. Cancellation changes actual state counts but is not used to justify the worst-case bound.

Theorem 3.4 (Fixed-color transfer complexity). *For a supplied crossing order of width w and fixed integer q , the one-tensor Potts transfer computes the partition function in*

$$n \operatorname{poly}(w, \log n) q^{\lfloor w/2 \rfloor + 2}$$

coefficient-ring operations and key-management work, and uses $\operatorname{poly}(w, \log n) q^{\lfloor w/2 \rfloor}$ coefficient slots and key storage. Preparation also uses $O(n)$ storage; coefficient bit memory is treated separately below. This replaces the general factorial state-count envelope by a single-exponential upper bound, without asserting that the baseline always realizes its larger envelope.

Proof. After a step there are at most $B_q(f_i) \leq q^{w/2}$ keys. Introducing at most two vertices has at most q^2 labeled choices per state; orbit aggregation cannot increase that count. Edge multiplication and forgetting use polynomial work in the key length. Sum over n edges. Two successive tables and a bounded extension list suffice. A deterministic comparison-based dictionary adds a polynomial factor in w and $\log n$; it does not change the exponential term. $\square$

The actual Python implementation uses dictionaries. Its timings concern that implementation, while the deterministic bound above can be realized with balanced search structures or sorted merging. Order construction is separate: the repository’s heuristic is polynomial work and is timed explicitly in end-to-end measurements. No optimal-width order is assumed to be available for free.

Corollary 3.5. *For fixed q , the scalar filter is polynomial time on supplied orders with $w = O(\log n)$ and runs in $n^{O(\log n)}$ time on supplied orders with $w = O((\log n)^2)$, including the bit costs of Section 4.*

The preceding matching count gives $2^{O(w \log w)}$ as a general envelope. At $w = O((\log n)^2)$ that is still quasi-polynomial in the broad definition, but contains an additional $\log \log n$ in the exponent. The sharper Potts bound should not be described as having changed a non-quasi-polynomial class into a quasi-polynomial one at that same width.

4 Division-free quadratic arithmetic and bit complexity

4.1 An exact ring for each integer color count

Fix an integer $q \geq 5$ and put

$$R_q = \mathbb{Z}[x]/(x^2 - (q-2)x + 1).$$

The discriminant $(q-2)^2 - 4$ lies strictly between the consecutive squares $(q-3)^2$ and $(q-2)^2$, so the polynomial is irreducible over $\mathbb{Q}$. In either real embedding,

$$x + x^{-1} = q - 2, \quad x^{-1} = q - 2 - x, \quad (x + 1)^2 = qx.$$

In particular x is a unit. Each element has a unique representation $a + bx$ with $a, b \in \mathbb{Z}$. Pair inequality is exact inequality of the algebraic numbers; it is not a numerical tolerance test.

For $q = 6$, the positive roots are $2 + \sqrt{3}$ and $2 - \sqrt{3}$. The Jones variable in our derivation is $t = A^{-4} = x^{-1}$. Changing embeddings interchanges these reciprocal values; equality to 1 is preserved because the coefficients are rational integers. The code works in the ring and need not select a real embedding.

Operation	Pair formula
Addition	$(a, b) + (c, d) = (a + c, b + d)$
Multiplication	$(a, b)(c, d) = (ac - bd, ad + bc + (q - 2)bd)$
Multiply by $-x$	$(a, b) \mapsto (b, -a - (q - 2)b)$
Multiply by $-x^{-1}$	$(a, b) \mapsto (-(q - 2)a - b, a)$
Inverse of x	$x^{-1} = (q - 2, -1)$

Table 2: Exact arithmetic used by the transfer. Full pair multiplication is needed for normalization and component merges; a single-edge transfer uses only additions and small integer multiples.

Equations (4) and (6) now define a division-free one-tensor algorithm. Negative powers in normalization are powers of the explicit unit x^{-1} . Fast powering is exact. No prime choice, probability-of-error estimate, or floating-point precision policy is required.

Proposition 4.1 (Sound exact obstruction). *If the partition function and Z_U are unequal as pairs in R_q , then D is a nontrivial knot. If they are equal, the computation gives no unknot certificate.*

Proof. The nonzero normalization factor in (5) identifies partition equality with equality of the selected normalized Jones value to 1. Every unknot has normalized Jones polynomial 1, so inequality rules it out. The converse would require an additional detection theorem that is not available here. Section 6 gives explicit failures at $q = 5$. $\square$

4.2 Coefficient lengths

Let $\|a + bx\|_1 = |a| + |b|$. The two equal-spin multiplication maps in Table 2 have induced ℓ^1 norm at most $q - 1$.

Theorem 4.2 (Linear coefficient bit lengths). *After i edges and I introduced vertices, every aggregate table coefficient, including every partial sum inserted into the next table, satisfies*

$$\|a + bx\|_1 \leq q^I (q - 1)^i \leq q^v (q - 1)^n.$$

Its magnitude coefficients therefore have at most

$$\lfloor v \log_2 q + n \log_2 (q - 1) \rfloor + 1$$

bits. For fixed q this is $O(n)$, and for $q = 6$ the simpler upper bound $5n + 4$ is valid.

Proof. Every labeled assignment of the introduced vertices contributes a product of i edge weights. Its norm is at most $(q - 1)^i$. An aggregate coefficient is a sum over a subset of at most q^I assignments, so the triangle inequality proves the first bound. Partial insertions are sums over subsets of the same histories and obey the same bound. The connected Tait graph has $v \leq n + 1$, giving the stated bit length. For $q = 6$, $\log_2 6 + \log_2 5 = \log_2 30 < 5$ and $\log_2 6 < 3$, proving the convenient integer bound. $\square$

The normalization pair is also polynomial in bit length:

$$\|Z_U\|_1 \leq (q - 1)^{|h|} q^{v+1}, \quad |h| \leq n + (v + 1)/2.$$

Thus no hidden exponential coefficient size invalidates the transfer's width bound. At fixed q , a single-edge update costs $O(n)$ elementary bit operations apart from key management. Ordinary multiplication of $O(n)$ -bit integers suffices for powering and component merges; faster integer multiplication can be used without changing the width dependence.

The coefficient-length bound is a worst-case upper bound, not an assertion that every table entry reaches it. The experiment record reports the largest coefficient actually inserted. It does not treat Python object overhead as coefficient bits, and neither number is a process-memory limit.

4.3 The modular comparison backend

For a second implementation path the package includes a modular $q = 5$ backend over $p = 2^{61} - 1$, with a root of $x^2 - 3x + 1$ and a compatible fourth root A . It compares the full bracket at precisely that A with $\delta(-A^3)^{\text{wr}(D)}$. The constants are checked algebraically on import. The old matching transfer can be evaluated at the same A , making an apples-to-apples scalar comparison possible.

A modular mismatch is also a sound obstruction. A modular match may arise from reduction modulo p or from a genuine specialization collision. Increasing the number of primes can address the former but cannot repair the latter. In particular it cannot repair the family in Section 6. The exact backend makes this distinction observable.

5 Factoring intermediate components instead of correlating all spins

5.1 Why the global orbit table can still be wasteful

A poor crossing order can create several disjoint components of processed Tait edges. Before any processed edge joins them, their partial weights factor. A global S_q orbit table nevertheless records whether an active spin in one component happens to have the same color as a spin in another. Those relative color identities matter only when a future edge first connects the two components. Recording them earlier can create many keys with an exactly factorizable coefficient.

This is an exact independence statement about the processed interaction graph. It does not follow merely because two groups of vertices appear far apart in a drawing, nor does it assume a graph separator whose crossing interactions have been discarded. We keep one tensor for each connected component of the graph formed by introduced vertices and processed edges. Connectivity through a forgotten vertex remains connectivity: forgetting does not undo an interaction already summed into the tensor.

An isolated newly introduced vertex has the one-key table $\{(0) \mapsto q\}$. An edge internal to a component uses the usual equality weight and forgetting operation. When no active vertices remain in a component, its scalar partition factor is multiplied into the accumulated scalar and its tensor is released. Only an edge joining two distinct processed components requires a new operation.

5.2 The relative-color merge

Let a left pattern have k classes and a right pattern have ℓ classes. Within each pattern, distinct classes must receive distinct colors. Across the two patterns, an arbitrary partial bijection may identify some left and right classes. If j pairs are identified, the merged pattern has $K = k + \ell - j$ classes, and it is possible exactly when $K \leq q$. The number of such alignments is

$$A_q(k, \ell) = \sum_{\substack{0 \leq j \leq \min(k, \ell) \\ k + \ell - j \leq q}} \binom{k}{j} \binom{\ell}{j} j!. \quad (8)$$

It is at most $q!$ when $k, \ell \leq q$. One way to see this bound is to fix distinct actual colors on all left classes and inject the alignment choices into the injective assignments of colors to the ℓ right classes by assigning fresh colors in a prescribed order. There are $(q)_\ell \leq q!$ such assignments. The implementation enumerates alignments directly, giving fresh classes their next canonical label.

Theorem 5.1 (Aggregate component merge). *Work over R_q , or over a field of characteristic zero or greater than q . Suppose the left and right orbit coefficients are C_L and C_R . For one relative alignment with K total classes, its aggregate coefficient before the joining edge is applied is*

$$\frac{C_L}{(q)_k} \frac{C_R}{(q)_\ell} (q)_{\underline{K}}. \quad (9)$$

After multiplication by the joining-edge weight, ordinary forgetting and canonical aggregation produce the correct merged tensor.

Proof. The orbit of a pattern with k nonempty classes has $(q)_k$ labeled active assignments. Color relabeling preserves the sum over forgotten spins, so all those active assignments have the same partial value. That value is $C_L/(q)_k$. The same statement holds on the right.

Before the joining edge, the two processed components share no vertex or processed interaction, so their per-assignment partial values multiply. The aligned combined pattern has $(q)_K$ labeled assignments. Multiplying by this orbit size gives (9). Every combined pattern determines exactly its restrictions and its partial-bijection alignment, so no term is omitted or counted twice. The joining edge and restriction to surviving vertices then obey Proposition 3.3. $\square$

In the modular backend, $q < p$ makes every falling factorial a unit in $\mathbb{F}_p$, and inverses can be precomputed. In the exact backend, each coefficient is an integral multiple of its orbit size in R_q . Both integer coordinates are therefore divisible by that positive integer. The implementation checks the remainders and performs exact integer division. It then multiplies the resulting pairs. This operation uses no fraction field and no approximate division. Unlike the one-tensor transfer, it is not literally division-free; the divisions are certified divisions by known small integers at fixed q .

5.3 A stronger parameter: the largest processed component

For each step, consider each still-active processed component and count its active vertices. Define g to be the largest such count over all components and all *post-edge* steps. Thus $g \leq \max_i f_i \leq w/2$.

Theorem 5.2 (Component-frontier complexity). *At fixed q , the factored transfer uses*

$$n \text{ poly}(g, \log n) q! q^{g+2}$$

coefficient operations and key-management work, with $O(nq^g)$ logical table keys. Over R_q , coefficient bit lengths remain $O(n \log q)$; the bit-operation bound is obtained by multiplying the displayed work by a polynomial in $n \log q$.

Proof. An edge can cause only its two incident vertices to be forgotten: any vertex whose last incident edge is the current edge must be an endpoint. Therefore a component's pre-forget active set at this step has at most $g + 2$ vertices. For a merge, this set is the union of the two input active sets, including newly introduced endpoints. If their sizes are a, b , then $a + b \leq g + 2$. The number of pairs of input keys is at most $q^a q^b \leq q^{g+2}$; each pair has at most $q!$ alignments. An internal edge has the same bound without the alignment factor. There are n edges and at most $O(n)$ active components.

For the exact coefficient bound, each component coefficient is a sum over assignments on a subset of vertices and a product over its processed edges. The argument of Theorem 4.2 applies componentwise. During a merge, the division removes an orbit multiplicity, and the final insertion is a sum over a subset of assignments on the union. Its length is again bounded by the same global $q^v(q - 1)^n$ envelope. $\square$

Bounding each input component separately by g and then multiplying two q^g tables would give the weaker exponent $2g$. The fact that only the current edge's endpoints can be forgotten is what gives the stronger $g + 2$ bound. A measurement of the largest table size alone would not prove either complexity result: merge Cartesian products and alignment work must be counted.

5.4 Implementation and remaining limitations

The package supplies modular and exact factored kernels. Their state budget bounds the sum of represented keys over current component tables. During an update, the old input tables and their replacement may coexist; normalized merge lists and cached alignment patterns also occupy memory.

Consequently that counter is a logical state budget, not an allocation or resident-memory ceiling. Transition counts include each relative alignment that is actually processed.

At $q = 5$ there are at most 120 alignments per input-key pair; at $q = 6$ there are at most 720. These constants can make factorization slower when components repeatedly merge without having saved many keys. The measured negative controls in Section 8 are therefore essential. The implementation does not factor a connected tensor merely because its coefficient array happens to have low rank, and it does not split components again when active graph connectivity changes. Such extensions require additional algebra and are research topics rather than current guarantees.

6 An infinite exact failure at five colors and its six-color repair

6.1 The weaving family as an adversarial test

Let

$$W_m = (\widehat{\sigma_1 \sigma_2^{-1}})^m.$$

The strand permutation is a power of a three-cycle, so W_m is a knot exactly when $\gcd(m, 3) = 1$. Its displayed diagram has $2m$ crossings and writhe zero. A familiar three-braid identity expresses its Jones polynomial using reduced Burau matrices [1, 3]:

$$V_{W_m}(t) = t + t^{-1} + \text{tr}(M(t)^m), \quad M(t) = \begin{pmatrix} 1-t & -t^{-1} \\ 1 & -t^{-1} \end{pmatrix}. \quad (10)$$

Direct calculation gives

$$\det M(t) = 1, \quad \text{tr } M(t) = 1 - t - t^{-1}.$$

Thus $T_m = \text{tr}(M(t)^m)$ obeys

$$T_0 = 2, \quad T_1 = 1 - t - t^{-1}, \quad T_{m+1} = (1 - t - t^{-1})T_m - T_{m-1}.$$

Only this elementary recurrence is needed for the specialization audit.

Theorem 6.1 (An exact five-color blind family). *At either root of $t^2 - 3t + 1 = 0$,*

$$V_{W_m}(t) = 3 + 2(-1)^m.$$

In particular every odd $m \geq 5$ with $3 \nmid m$ gives a nontrivial knot whose exact five-color Jones value equals the unknot value.

Proof. At this specialization, $t + t^{-1} = 3$ and $\text{tr } M = -2$. The trace recurrence has solution $T_m = 2(-1)^m$. Equation (10) gives the displayed value. Nontriviality follows either from Theorem 6.2 below or from the determinant formula

$$\det(W_m) = L_{2m} - 2 = \left(\frac{3 + \sqrt{5}}{2}\right)^m + \left(\frac{3 - \sqrt{5}}{2}\right)^m - 2 > 1$$

for $m \geq 2$ [1]. Here L_j denotes the Lucas sequence. For example W_5 has determinant 121. $\square$

This is a collision in a characteristic-zero number field. Both embeddings have the same value, and every finite-field image in which the specialization is defined inherits the equality. More precision, a conjugate root, or more primes at the same $q = 5$ point cannot solve it. The statement is a direct consequence of a classical trace identity; we make no claim that this particular consequence has not appeared elsewhere. Its role here is to disprove an overly optimistic completeness inference and to supply an infinite regression family.

6.2 Why six colors changes this family

Theorem 6.2 (Every exact $q \geq 6$ detects the nontrivial weaving members). *Let $q \geq 6$ be an integer and let $t + t^{-1} = q - 2$. For every $m \geq 2$ such that W_m is a knot, $V_{W_m}(t) \neq 1$.*

Proof. Put $a = q - 3 \geq 3$ and define

$$U_0 = 2, \quad U_1 = a, \quad U_{m+1} = aU_m - U_{m-1}.$$

The trace recurrence gives $T_m = (-1)^m U_m$, hence

$$V_{W_m}(t) = a + 1 + (-1)^m U_m. \tag{11}$$

We have $U_2 = a^2 - 2 > a$. If $U_m > U_{m-1} > 0$, then $U_{m+1} > (a - 1)U_m \geq 2U_m$. Thus U_m is positive and strictly increasing for $m \geq 1$. For even $m \geq 2$, (11) is greater than 1. For odd $m \geq 3$, $U_m > a$ and the expression is less than 1. The remaining member W_1 has value 1 and is the unknot. $\square$

At $q = 6$, the recurrence starts 2, 3, 7, 18, 47, 123, $\dots$. Thus $V_{W_5} = 4 - 123 = -119$ and $V_{W_7} = 4 - 843 = -839$. These integer values in a quadratic specialization provide especially simple independent checks of normalization and color count.

m	Crossings	Knot?	Exact $q = 5$ value	Exact $q = 6$ value
1	2	yes	1	1
2	4	yes	5	11
4	8	yes	5	51
5	10	yes	1	-119
7	14	yes	1	-839
8	16	yes	5	2211

Table 3: Weaving specialization values, computed from the displayed trace recurrence and independently checked by the transfer routines.

The equality-pattern spaces for $q = 5$ and $q = 6$ are identical for $f \leq 5$. At $f = 6$ their sizes are 202 and 203, respectively. On small frontiers, changing the specialization can therefore improve detection with little change in represented state count. Its worst-case exponential base does increase, so this observation is not a universal runtime comparison.

6.3 What this result does and does not decide

The exact backend defaults to $q = 6$ because the $q = 5$ family is a proved structural weakness. A fixed $q = 6$ equality is still inconclusive. Proving that $V_K(2 + \sqrt{3}) = 1$ forces the unknot for all knots would imply the usual Jones unknot-detection conjecture; no such theorem is supplied or assumed.

Moreover, the baseline’s short-braid stage already decides these weaving examples from their braid representation. Their value here is an adversarial test of a *filter*, not a newly solved class for end-to-end recognition. In isolated filter benchmarks that earlier stage is deliberately bypassed and the bypass is reported. Normal-pipeline benchmarks retain it.

7 Implementation, integration, and certificate semantics

7.1 The implemented backends

The Python package adds four core modules: a modular one-tensor transfer, an exact one-tensor transfer, a modular component transfer, and an exact component transfer. The first two provide independent execution paths for the same $q = 5$ specialization when the exact evaluator is configured with $q = 5$. The exact implementations support every integer $q \geq 5$ and default to $q = 6$.

Backend/API	Arithmetic	Role
matching	Existing finite field	Unchanged default Jones filter and same-point comparison reference.
potts5	$\mathbb{F}_{2^{61}-1}$	Opt-in five-color equality-pattern filter.
potts-exact	R_q , default $q = 6$	Opt-in exact one-tensor filter; simplest exact replay path.
potts-exact-factorized	R_q , default $q = 6$	Opt-in exact component filter, using integral orbit-size division.
factorized_potts_bracket	$\mathbb{F}_{2^{61}-1}$	Modular factored API, included as an experimental comparison kernel.

Table 4: Interfaces available in the delivered source tree. The backend selector here controls only the Jones stage; the Khovanov backend is a separate option.

The default remains the existing matching filter. This is a deliberate consequence of the evidence: the Potts encodings improve some inputs, but can regress on others, and $q = 5$ changes the invariant point in a way that misses a known family. An automatic policy should be supported by a broader cost model before changing existing behavior.

7.2 One-tensor pseudocode

Listing 1: Exact equality-pattern transfer at a fixed color count, for $n > 0$.

```

validate the knot diagram and supplied crossing permutation
choose one checkerboard color from the two complete profiles
compute first and last incident-edge positions for each Tait vertex
table = { empty_pattern: (1, 0) }
for edge in crossing_order:
    introduce its previously unseen endpoints
    following = {}
    for pattern, coefficient in table:
        for extension, multiplicity in color_orbit_extensions(pattern):
            value = coefficient * multiplicity
            if the endpoint classes agree:
                value *= -x^(-edge.omitted_exponent)
            target = restrict_and_canonicalize(extension, surviving_vertices)
            following[target] += value
            remove target immediately if its coefficient is exactly zero
            enforce state and transition budgets
    table = following
    forget vertices whose last edge has just been processed
Z, ZU = final partition pair, exact unknot normalization pair
return inequality as a KNOTTED obstruction; equality is inconclusive

```

The factored variant replaces the single table by a map from union-find component representatives to active lists and tables. It applies Theorem 5.1 when the current edge joins two representatives. The algebraic scalar associated with already closed components is retained.

7.3 Budget behavior and sound control flow

Both new exact backends accept a state budget, a transition budget, a cooperative deadline callback, an explicit crossing order, and an optional fixed shading. The enclosing recognizer computes an order once and reuses it for the Jones and possible Khovanov stages. Order construction itself checks the deadline. Local budget exceptions are caught as a skipped filter. A global time exception propagates to the existing resource-limit outcome.

The standalone `jones` command now handles resource exhaustion explicitly. Its JSON output is inconclusive and its exit code is 3. A completed equality is also inconclusive, but is a successful computation and exits 0. An invalid option or input exits 2. In particular, a truthy dictionary describing a skipped computation must never be interpreted as a knottedness witness. Focused tests exercise this boundary, including zero budgets on actual unknots and exact $q = 5$ equality on a nontrivial knot.

All verdicts concern validated classical knot input. The low-level `Diagram(pd)` constructor is used internally for already validated data, including benchmark repetitions. Public file reading and certificate replay use the validating constructors. Neither a genus error in a rotation system nor a multiple-component input is silently treated as an ordinary knot.

7.4 Exact values, serialization, and replay

The raw exact evaluators return integer coordinate pairs, along with the chosen shading, normalization exponent, writhe, frontier counts and work counters. Obstruction wrappers encode the two large coordinate pairs as canonical signed hexadecimal strings: `partition_function_hex` and `unknot_partition_hex`. This permits arbitrarily long bounded-width examples to be serialized without hitting Python’s limit on decimal conversion of huge integers. It does not change a process-global interpreter setting or truncate values.

The current replay schema is restricted to $q = 6$, for either exact backend; the raw evaluation APIs support all $q \geq 5$. The certificate format binds a complete PD, an explicit crossing permutation, the chosen shading, and the full exact evidence record. Its SHA-256 input digest detects accidental edits; it is not a digital signature. The replay tool validates the diagram, recomputes the chosen exact transfer, compares all evidence fields, and independently checks that the two coordinate pairs differ. It does not trust a stored Boolean flag. Tampering with the order, normalization, values, flags or counters is rejected.

This is *recomputation-based replay*, not a separate polynomial-size proof checker whose complexity is independent of the original transfer. Its runtime has the same width dependence. The independent smoothing-cube oracle is a different algorithm and supplies cross-validation on smaller inputs; it is not the large-instance replay implementation. If a verifier’s resource budget is exhausted, its result is `UNVERIFIED`, not a positive or negative topological conclusion.

7.5 Command examples

Run these commands from the included `fast` directory:

```
python -m fastunknot jones examples/conway.json \
    --backend potts-exact --potts-colors 6

python -m fastunknot recognize examples/conway.json \
    --jones-backend potts-exact-factorized --potts-colors 6

python -m fastunknot jones examples/hard_unknot_8.json \
    --backend potts-exact --max-transitions 0
```

The second command retains all earlier recognition stages, so its winning method may precede the Jones stage. To study the scalar algorithm in isolation, use the benchmark scripts or the low-level APIs with an explicitly recorded crossing order.

8 Verification and controlled computational results

8.1 Independent correctness checks

The integrated repository suite passed **172 tests**, including all 134 baseline tests and 38 added Potts and integration tests. The baseline and modified logs are both included. Their elapsed suite times are not performance comparisons, since the modified suite performs additional work.

The strongest cross-check constructs the entire normalized Jones Laurent polynomial directly from each smoothing cube. It uses union-find on edge labels, independent state-circle enumeration, and a separate quadratic evaluation recurrence. It does not construct Tait graphs or use a frontier transfer to obtain its expected answer.

Independent or integration check	Completed count
Validated diagrams in the full-polynomial oracle	248
Complete smoothing states enumerated	30,116
One-tensor exact color/shading/order comparisons	4,464
Factored exact color/shading/order comparisons	4,464
Additional weaving trace comparisons, each exact backend	45
Adversarial direct labeled-spin comparisons with cancellations	432
Residue annihilator and Fitting/Drazin identity cases	524
Closed inverse-formula checks	108
Radical-cutoff checks	173
General- p sharpness witnesses	126
Nilpotent-residue companion witnesses	18
Accepted certificate kinds in replay self-test	2
Rejected tampered certificates	21

Table 5: Verification counts from the saved execution records. These finite checks supplement the proofs; they do not replace them.

The polynomial comparisons cover $q = 5, 6, 7$, both checkerboard colors, and natural, reversed and random crossing orders. Fixtures include Conway, Kinoshita–Terasaka, a hard unknot, and connected sums. The additional weaving comparisons extend to $m = 49$, or 98 crossings. Separate

focused tests cover all 24 crossing orders of the figure-eight diagram, mirrors, loops, parallel edges, orbit alignment cardinalities, invalid options, deadline propagation and exact budget boundaries.

The radical-ring oracle uses exponent tuples rather than the production packed binary algebra. Its 524 annihilator/Fitting cases include every 2×2 matrix over $\mathbb{F}_2[x]/(x^2)$, a set of 256 matrices. The sharpness checks vary p, b, r and test both the surviving preceding power and the first forced zero power. A separate finite-menu oracle checks 243 Laurent polynomials, 20 products of quadratic factors, and 5,151 writhe-dependent support intervals for Proposition 10.1.

Finally, a real exact certificate for W_{6001} , with 12,002 crossings, was serialized and replayed. Its largest coordinate had 21,791 bits. The interpreter’s decimal-conversion limit remained unchanged. This tests the hexadecimal evidence format on actual large algebraic output; it is not a new end-to-end recognition result for a short-braid family.

8.2 Kernel benchmark design

The principal timing record contains **3,045 timed calls**: 29 cases, 15 rounds and seven arms. The environment was Python 3.12.14 on Linux 6.18.44 with glibc 2.39. All raw nanosecond samples, arm orders, inputs, crossing orders, counters and source hashes are included. The corpus was specified as follows:

1. all 14 named JSON fixtures in the pinned implementation, with the existing greedy order;
2. the first 12 accepted one-component five-strand, 16-crossing random braid diagrams and their shuffled orders from seed 764259;
3. the three known five-color collision controls W_5, W_7, W_{11} .

No random case was selected by its measured speedup. The deliberately poor orders test representation sensitivity; they are not claimed to approximate an input distribution in applications.

The seven arms are matching A1, modular $q = 5$, factored modular $q = 5$, exact $q = 5$, exact $q = 6$, factored exact $q = 6$, and matching A2. The matching arms use the same modular A as the five-color Potts specialization. The exact six-color arms evaluate a different point and have different collision behavior. They are valid cost comparisons, not same-value numerical comparisons.

Each timed call receives the same supplied order and a fresh already validated diagram object. Imports, input validation, order search and diagram construction are outside the timed region. Tait extraction, shading selection, transfer and normalization are inside it. The counters and output values must reproduce exactly across repetitions. Arm positions rotate between rounds to reduce systematic ordering effects. The two identical matching arms provide an A/A noise control.

For each new arm, we report the median of the paired ratios $T_{\text{matching A1}}/T_{\text{arm}}$. It need not equal the ratio of the separately reported median times. The saved order-statistic interval has nominal coverage at least 95 percent under independent identically distributed rounds. Shared-machine timing noise and dependence between rounds limit that interpretation; these are descriptive intervals, not universal performance guarantees. A local faster/slower label additionally requires the median ratio to lie beyond the corresponding A/A interquartile range. No multiple-comparison or population-level conclusion is drawn.

8.3 Measured improvements and regressions

The clearest factorization result is `random_order_05`. The exact one-tensor implementation reaches 4,111 represented keys and 30,197 transitions. Its factored counterpart reaches 205 total represented

Input	n	w	f	g	K_E	K_F	T_M/T_E	T_M/T_F
Conway	11	6	3	3	5	6	1.574 [†]	1.298
Kinoshita–Terasaka	11	6	3	3	5	6	1.286 [†]	1.048
Conway #8	88	6	3	3	5	6	1.704 [†]	1.215
Random 04	16	22	8	7	4,111	317	0.118 [†]	0.376 [†]
Random 05	16	26	8	6	4,111	205	0.083 [†]	2.054 [†]
Random 11	16	22	4	4	15	15	11.220 [†]	11.855 [†]

M : matching kernel at the five-color evaluation point; E : exact $q = 6$; F : factored exact $q = 6$. K_E, K_F count peak represented keys; K_F sums represented component tables. f is the full active spin frontier and g the largest active frontier of one processed component. Ratios are paired medians; values above one favor the denominator. [†] marks a comparison meeting both the pointwise interval and A/A rule.

Table 6: Selected supplied-order kernel results. The companion CSV retains all 29 cases, including every measured regression.

keys, at most 203 in one component, and 1,558 transitions. The maximum global active-face count is eight and the maximum component frontier is six. These discrete counts reproduce independently of timing noise.

On this input, the paired one-tensor/factored exact median ratio is 23.843, with recorded interval [19.448, 29.683]. The same-specialization comparison is the strongest evidence for the component optimization. Against the matching arm, factored exact $q = 6$ has median ratio 2.054 and interval [1.878, 2.552].

The negative control `random_order_04` is equally informative. Factorization improves its own exact backend by a median factor of 3.186, yet its ratio against matching is only 0.376: matching remains substantially faster. Several other arbitrary orders also favor matching. The factorized $q = 6$ arm has five locally faster cases, nine slower cases and fifteen without a clear timing claim under the recorded rule. The one-tensor exact $q = 6$ arm has eight, eight and thirteen, respectively.

Several larger named fixtures show favorable kernel medians, but many intervals and A/A ranges are too wide to support a local claim for a particular arm. These cases are retained without upgrading the evidence. The examples establish that the representation can improve practical work; they do not establish a growing-family runtime exponent.

8.4 Normal-pipeline experiment

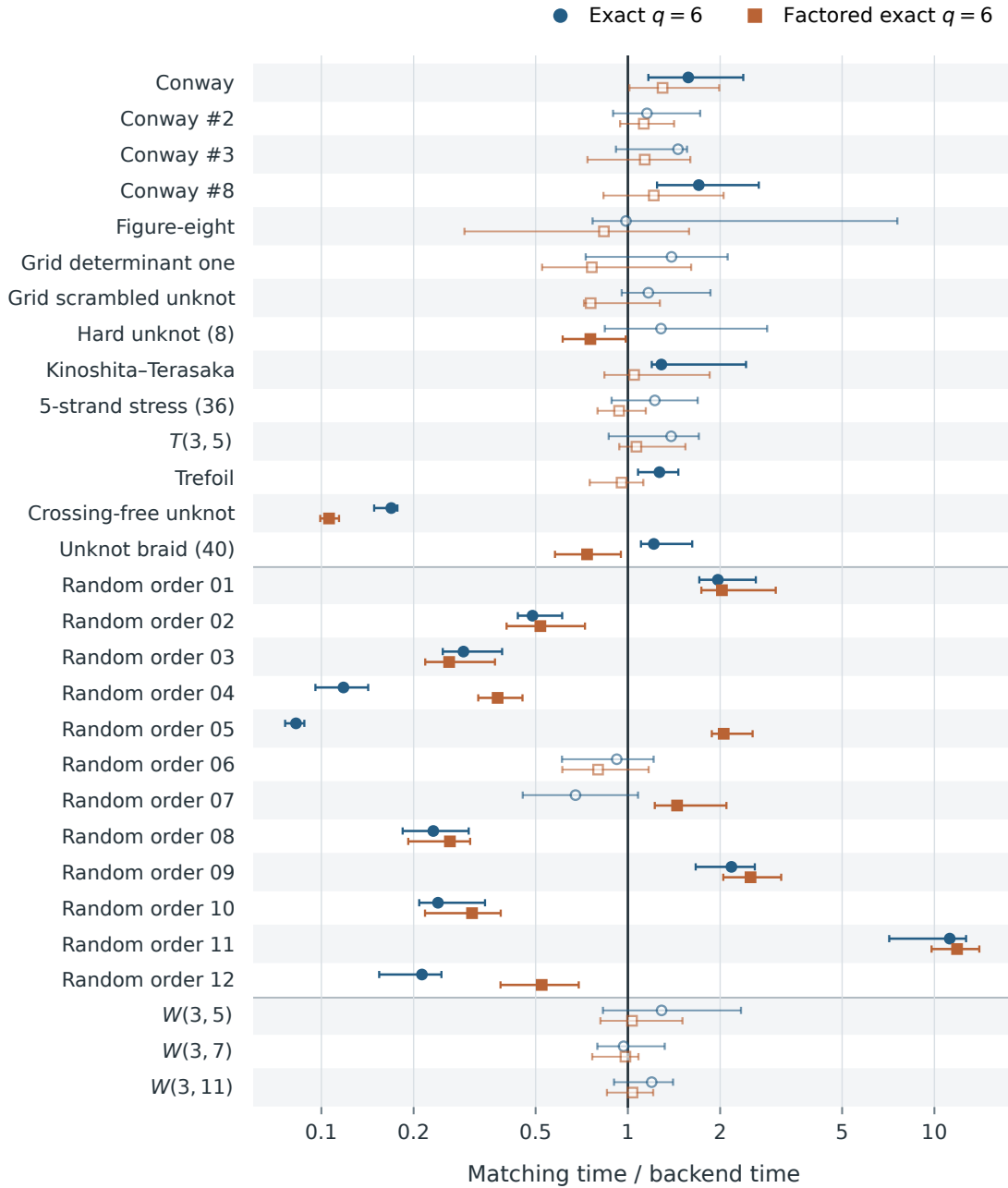
A separate experiment contains **476 timed calls**: 17 inputs, seven rotated rounds, matching A/A controls, exact $q = 6$, and factored exact $q = 6$. It includes JSON parsing, validating diagram construction, normal recognition and JSON output encoding. Imports, file I/O, CLI parsing and process startup are excluded. The matching backend uses its native generic A , and every ordinary early stage remains enabled.

All verdicts are complete and consistent. Only five cases reach Jones: Conway, Kinoshita–Terasaka, and Conway connected sums with two, three and eight summands. The other twelve are decided earlier, including all three weaving controls.

No full-pipeline speedup is supported by this seven-round run. The five cases that exercise Jones have wide intervals or A/A variation. Backend speed claims are explicitly suppressed on bypassed cases. This result limits the practical conclusion without weakening the proved transfer bounds or the deterministic resource savings.

Paired filter-kernel timings

All 29 inputs · 15 rounds · supplied crossing orders



Filled: interval + A/A rule met; hollow: no clear claim. Bars: nominal pointwise intervals.

Matching uses the $q=5$ point; exact backends use $q=6$. Ratios above 1 favor Potts.

Figure 1: All 29 kernel comparisons against the same modular matching reference. Ratios above one favor the indicated Potts arm. Exact $q = 6$ uses a different Jones point from the modular reference. Intervals and claim markers follow the recorded paired/A/A rule; regressions are shown alongside improvements.

Resources across the random-order controls

Exact six-color arithmetic · all 12 accepted controls from seed 764259

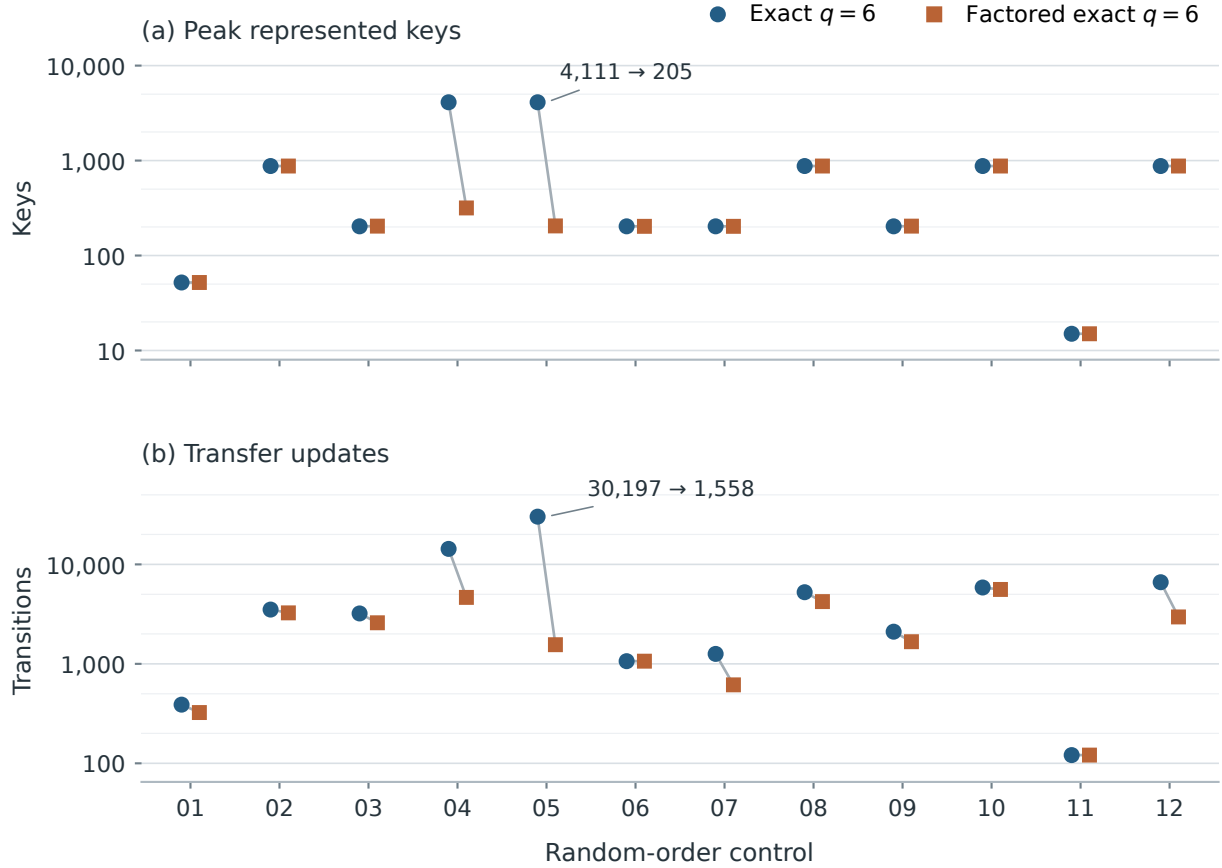


Figure 2: Logical state and transition counts on all twelve fixed random-order controls. These deterministic measures expose both the savings from component independence and cases where little factorization is available. State counts are not process-memory measurements.

Input	n	T_M/T_E [interval]	T_M/T_F [interval]	Supported claim
Conway	11	0.991 [0.448, 3.221]	1.717 [0.605, 3.925]	None
Conway #2	22	1.175 [0.582, 4.046]	1.159 [0.286, 4.324]	None
Conway #3	33	1.202 [0.292, 2.032]	0.873 [0.426, 2.014]	None
Conway #8	88	1.645 [0.474, 4.439]	0.906 [0.346, 1.539]	None
Kinoshita-Terasaka	11	1.374 [0.435, 3.422]	1.973 [0.305, 3.946]	None

M : the normal pipeline with its native generic matching evaluation; E, F : the normal pipeline with exact or factored exact $q = 6$. Nominal pointwise intervals; see the experimental-design assumptions in the text. All five inputs reached Jones evaluation; the other twelve inputs bypassed it. No comparison in this seven-round pipeline run met the timing-claim rule.

Table 7: The five normal-pipeline cases that exercised the selected Jones backend. Timings include input parsing, checked diagram conversion, recognition, and JSON output encoding in a hot process.

8.5 Provenance and reproducibility limits

The kernel source hashes agree before and after its timing run. Subsequent edits changed only exact obstruction serialization wrappers and added their shared helper; every raw evaluator and helper used in the kernel timing has unchanged AST content. The exact measured versions of all eight relevant modules were reconstructed and checked against the recorded complete-file SHA-256 hashes. They are included as a provenance snapshot, with the wrapper-only patch and audit record. The normal-pipeline measurement uses the final wrappers and integrated backends, with its own matching before/after hashes.

The corpus is small and timings are machine-dependent. Default budgets were raised consistently for the kernel experiment to avoid comparing early exits; all completed results and detection outcomes were checked. There is no evidence here that a single representation dominates every diagram, that an adaptive policy has already been validated, or that the general recognizer has become quasi-polynomial.

9 Residue characteristic polynomials and short algebraic certificates

The Potts improvements act on a scalar obstruction. A separate line of progress concerns the local rings that occur in exact categorical Khovanov computation. Earlier ProveIt work already uses nilpotent radical perturbations and finite geometric-series inverses [17, 20]. The immediately preceding continuation constructs Fitting decompositions of supplied *scalar* binary endomorphisms. We now extend the algebraic certificate mechanism to supplied ring-valued, same-matching blocks.

The ingredients are standard Cayley–Hamilton, scalar Frobenius, polynomial Chinese remainders, and Fitting decomposition. The theorem is stated with a complete elementary proof because the applicable ring and matrix hypotheses matter. This is not a literature-wide novelty claim about matrix nil indices. No production scanner speedup from this supplemental result is claimed.

9.1 A residue-only annihilator

Fix a prime p and positive integers b, r . Set

$$R_{b,p} = \mathbb{F}_p[x_1, \dots, x_b] / (x_1^p, \dots, x_b^p), \quad J = (x_1, \dots, x_b), \quad B = b(p-1).$$

Let $\rho : R_{b,p} \rightarrow \mathbb{F}_p$ take the constant coefficient. The monomial basis has exponents $0 \leq \alpha_i < p$, so $J^{B+1} = 0$ and $J^B \neq 0$. Scalar Frobenius gives

$$a^p = \rho(a) \quad (a \in R_{b,p}). \tag{12}$$

This identity concerns scalar elements of a commutative ring. The map $A \mapsto A^p$ on matrices is not entrywise Frobenius and need not be additive.

Theorem 9.1 (Residue characteristic-polynomial certificate). *For $A \in M_r(R_{b,p})$, put $A_0 = \rho(A)$ entrywise and $\chi_0(t) = \det(tI - A_0) \in \mathbb{F}_p[t]$. Then*

$$\boxed{\chi_0(A)^p = 0.}$$

Thus A has an explicitly computable annihilating polynomial $\chi_0(t)^p$ of degree pr , independent of b .

Proof. Write $\chi_A(t) = \sum_{i=0}^r c_i t^i$ over $R_{b,p}$, with $c_r = 1$. Taking constant coefficients commutes with the determinant, so $\chi_0(t) = \sum_i \rho(c_i) t^i$. Cayley–Hamilton over a commutative ring gives $\sum_i c_i A^i = 0$. These summands commute with each other: each is a scalar multiple of a power of the same matrix. Consequently raising their sum to its p th power is legitimate and yields

$$0 = \sum_i c_i^p A^{pi} = \sum_i \rho(c_i) A^{pi} = (\chi_0(t)^p)(A) = \chi_0(A)^p.$$

No Frobenius identity for arbitrary noncommuting matrix summands was used. $\square$

The practical case here is $p = 2$, where the dot algebra of a same-matching block has square-zero variables. Its annihilator degree is at most $2r$, even if the scalar coefficient space has dimension 2^b .

9.2 A sharp radical-matrix bound

Theorem 9.2 (Sharp universal radical cutoff). *Every $N \in M_r(J)$ satisfies*

$$N^L = 0, \quad L = \min\{b(p-1) + 1, pr\}.$$

For every $b, r \geq 1$ and prime p , there is such an N with $N^{L-1} \neq 0$.

Proof. The ideal bound gives $N^{B+1} = 0$. Since $N_0 = 0$, $\chi_0(t) = t^r$, and Theorem 9.1 gives $N^{pr} = 0$.

For sharpness, let $s = \min(b, r)$. Use an $s \times s$ upper-bidiagonal block with distinct variables $x_1, \dots, x_s$ on the diagonal. Of its $s-1$ superdiagonal positions, label

$$e = \min\{s-1, (b-s)(p-1)\}$$

with variables among $x_{s+1}, \dots, x_b$, each used at most $p-1$ times. Label the remaining positions with diagonal variables, again with each used at most $p-1$ times. There is enough capacity since $s-1 \leq s(p-1)$. Let a_i be the number of superdiagonal entries labeled x_i for $i \leq s$.

In a power of this bidiagonal matrix, a path from row 1 to column s traverses every superdiagonal position in its forced order and can wait at vertex i by using the diagonal x_i . Choose exactly $p-1-a_i$ waits at that vertex. The resulting monomial has degree

$$D = (s-1) + \sum_{i=1}^s (p-1-a_i) = s(p-1) + e = \min\{b(p-1), pr-1\} = L-1.$$

Every variable exponent is below p . Its coefficient is one: the superdiagonal order is forced and the distinct diagonal variables specify the waiting counts uniquely. Hence it survives in the $(1, s)$ entry of N^D . Embed the block into an $r \times r$ zero matrix when $s < r$. $\square$

For $p = 2$ the cutoff is $\min(b+1, 2r)$. When $b \geq 2r-1$, a particularly transparent witness has r distinct diagonal variables and $r-1$ further distinct superdiagonal variables. The $(1, r)$ entry of N^{2r-1} is their product. This shows that replacing $2r$ by r , based on a scalar square-zero intuition, would be wrong.

If instead A_0 is nilpotent of index s , then $A^s \in M_r(J)$. Combining the ideal bound with Theorem 9.1 yields

$$A^{\min\{pr, s[b(p-1)+1]\}} = 0. \tag{13}$$

The pr term can be attained already with one variable: the companion matrix of $t^r - x_1$ has $A^r = x_1 I$ and nilpotence index pr . We do not claim simultaneous sharpness of (13) for every prescribed triple (b, r, s) .

9.3 Inverses without a determinant over the large ring

Corollary 9.3. *For $N \in M_r(J)$,*

$$(I + N)^{-1} = \sum_{j=0}^{L-1} (-N)^j, \quad L = \min\{b(p-1) + 1, pr\}.$$

More generally, if A_0 is invertible and $\chi_0(t) = \sum_{i=0}^r c_i t^i$, then

$$A^{-1} = -c_0^{-1} \sum_{i=1}^r c_i A^{pi-1}.$$

Proof. The first formula telescopes using $N^L = 0$. For the second, Theorem 9.1 says $c_0 I + \sum_{i=1}^r c_i A^{pi} = 0$. Multiplying the proposed inverse on either side by A gives I . $\square$

At $p = 2$, the minus signs disappear and $c_0 = 1$. The characteristic polynomial is computed over the small base field, not by expanding a determinant with 2^b coefficients in each entry. The powers of A still require actual ring arithmetic. If a geometric series is evaluated by doubling a power-and-partial-sum pair, it takes $O(\log L)$ matrix products. The comparison with earlier methods must acknowledge this: reducing a cutoff from $b + 1$ to $2r$ does not automatically save b/r matrix products in an already doubled implementation.

9.4 Constructive Fitting and Drazin data

Factor the residue characteristic polynomial as

$$\chi_0(t) = t^a q_0(t), \quad q_0(0) \neq 0.$$

The notation q_0 here is a polynomial and is unrelated to a Potts color count. Polynomial CRT gives a residue class $e(t)$ modulo $t^{pa} q_0(t)^p$ satisfying

$$e(t) \equiv 0 \pmod{t^{pa}}, \quad e(t) \equiv 1 \pmod{q_0(t)^p}.$$

Choose its representative of degree below pr and set $E = e(A)$.

Theorem 9.4 (Explicit Fitting projectors). *The operators above satisfy*

$$E^2 = E, \quad AE = EA, \quad A^{pa}(I - E) = 0, \quad q_0(A)^p E = 0.$$

There is a polynomially specified D with $AD = DA = E$ and $DE = ED = D$. The modules $\ker E$ and $\operatorname{im} E$ are free of ranks a and $r - a$, and a concrete basis change can be lifted from residue projector columns.

Proof. All displayed identities for E follow from polynomial divisibility by $t^{pa} q_0^p$ and Theorem 9.1. Choose $z(t)$ by CRT with $z = 0$ modulo t^{pa} and $z = t^{-1}$ modulo q_0^p , and let $D = z(A)$. The claimed identities follow by the same argument. In particular A is nilpotent on $\ker E$ and invertible on $\operatorname{im} E$. The limiting cases $a = 0$ and $a = r$ use $E = I$ and $E = 0$ respectively.

Choose columns of E_0 spanning $\operatorname{im} E_0$ over $\mathbb{F}_p$, and columns of $I - E_0$ spanning $\ker E_0$. Take the corresponding columns of E and $I - E$ and arrange the kernel columns first in a matrix P . Its residue P_0 is invertible. Since $R_{b,p}$ is local with maximal ideal J , $\det P$ is a unit, so P is invertible. The column identities give $EP = P \operatorname{diag}(0_a, I_{r-a})$, proving the rank and basis claims. $\square$

The resulting D is the Fitting/Drazin inverse: $DAD = D$ and $A^{pa+1}D = A^{pa}$. The method constructs the decomposition inside matrices over $R_{b,p}$. It need not first expand the module to dimension rp^b over the base field. It nevertheless must represent and manipulate its ring coefficients.

9.5 Intertwining across categorical blocks

Suppose a finite additive $\mathbb{F}_p$ -linear category contains same-matching objects X_i with $\text{End}(X_i) = R_{b_i,p}$, and a degree-preserving chain endomorphism Q is block diagonal by degree and matching. Write A_i for its block on $X_i^{r_i}$. A differential component T satisfies $A_j T = T A_i$ under actual categorical composition.

Construct the respective E_i, D_i and E_j, D_j by the preceding theorem. Their projector polynomials need not be the same. For N large enough to kill both nilpotent restrictions,

$$(I - E_j)TE_i = (I - E_j)TA_i^N D_i^N = (I - E_j)A_j^N T D_i^N = 0.$$

Similarly,

$$E_j T(I - E_i) = D_j^N A_j^N T(I - E_i) = D_j^N T A_i^N (I - E_i) = 0.$$

Hence $E_j T = T E_i$, and the nilpotent and invertible images form complementary subcomplexes. The lifted bases exhibit a genuine chain isomorphism when every inverse and categorical cross term is evaluated.

The characteristic-polynomial theorem is used only inside the commutative same-matching endomorphism blocks. Applying it directly to an arbitrary mixed-matching matrix with noncommutative cobordism entries would be invalid. Finding useful Q , keeping it small, and exploiting the resulting summands are separate algorithmic problems.

9.6 How this advances the previous continuation

The previous scalar Fitting procedure already stabilizes in at most its matrix dimension over $\mathbb{F}_2$. The $2r$ annihilator does not accelerate that scalar computation. Its contribution is a ring-valued certificate whose polynomial degree is controlled by block multiplicity rather than only by the number of dot variables. The sharpness theorem prevents an unjustified further truncation.

The supplied oracle represents monomials by exponent tuples and checks the identities independently of the production dot algebra. Its characteristic polynomial routines are intended for small exhaustive experiments, not as a scalable implementation. The exact counts are reported in Section 8. No bound on all scanner multiplicities, all block sizes, or closure of a succinct representation follows from these local theorems.

10 The remaining distance to quasi-polynomial recognition

10.1 Three separate obstacles

A faster exact scalar filter addresses only one part of the recognition problem. Three distinct missing implications must not be conflated.

First, equality of a selected Jones value to 1 need not decide the knot. At $q = 5$ this is disproved by Theorem 6.1; at $q = 6$ we assume no global detection statement. The completeness of the enclosing recognizer comes from its retained exact fallback, not from these evaluations.

Second, the input diagram does not come with a promise of logarithmic or polylogarithmic width. Theorems 3.4 and 5.2 are bounds in a supplied order's measured parameters. A polynomial heuristic finds useful orders but has no universal $O((\log n)^2)$ guarantee. Planar separator methods and tensor contraction are relevant, established tools [11, 22]; using them does not imply a polylogarithmic bound for arbitrary diagrams.

Third, the full categorical complex contains more information than the scalar partition function. Boundary size does not alone control object multiplicities, ranks, differential supports or the size of an explicit homology basis. Kelomäki and Schütz prove exponential behavior for explicit scanning or divide-and-conquer constructions even on certain positive and alternating three-braid families, while giving polynomial algorithms for compressed three-braid homology and fixed homological windows at fixed strand count [5]. This is a representation obstruction, not a reason to abandon compression.

10.2 A finite exact menu decides Jones-polynomial triviality

The fixed-color obstruction can be extended to an unconditional decision about the entire Jones polynomial, although this still does not settle unknot recognition.

Proposition 10.1 (A growing exact specialization menu). *For a knot diagram with $n > 0$ crossings, suppose the exact quadratic evaluations equal 1 for every integer $q \in \{5, 6, \dots, 2n + 5\}$. Then $V_D(t)$ is identically 1. Conversely, the constant polynomial 1 passes every evaluation. On a supplied order of width w , this decision can be carried out using the one-tensor algorithm in*

$$n^{\lfloor w/2 \rfloor + O(1)}$$

bit operations. In particular it is quasi-polynomial when $w = O(\log n)$.

Proof. In a normalized smoothing summand, the A exponents come from a smoothing weight in $[-n, n]$, the writhe correction in $[-3n, 3n]$, and δ^{c-1} with $c - 1 \leq n$. Hence all A exponents lie in $[-6n, 6n]$. For a knot they are multiples of four, so all Jones t exponents lie in the conservative interval $[-2n, 2n]$. Thus $P(t) = t^{2n}(V_D(t) - 1)$ is a polynomial of degree at most $4n$.

Equality at a root of $t^2 - (q - 2)t + 1$ makes this irreducible quadratic divide P over $\mathbb{Q}$. The $2n + 1$ quadratics for the specified distinct values of q are pairwise coprime. If $P \neq 0$, their product of degree $4n + 2$ cannot divide it. Thus $P = 0$.

There are $O(n)$ evaluations with $q = O(n)$. For $f \leq w/2 \leq n$, the number of equality patterns is at most the Bell number $B(f) \leq f^f \leq n^f$, with the empty pattern treated separately. Introducing two vertices has only $O((f + 2)^2)$ canonical branches, independent of the multiplicities attached to those branches. The coordinate lengths are $O(n \log n)$ by Theorem 4.2. All remaining factors, including exact arithmetic and normalization, are polynomial in n and w . $\square$

A smaller menu follows from the same proof without new topology. For writhe r , the actual conservative exponent interval is $[\lfloor 3(r - n)/4 \rfloor, \lfloor 3(r + n)/4 \rfloor]$. It contains zero and has width $D \leq \lfloor 3n/2 \rfloor$. Thus $\lfloor D/2 \rfloor + 1$ distinct quadratic evaluations suffice. The larger $2n + 1$ menu keeps the statement independent of this refinement.

This is elementary degree counting, not a new claim about the difficulty of the Jones polynomial. It shows exactly what a sufficiently large menu can certify. The menu is a mathematical extension; the default recognizer does not automatically run $O(n)$ different color counts. The complete unknot-detection question for $V_D(t) \equiv 1$ remains outside its conclusion. It also shows why a growing number of colors cannot be hidden in a fixed- q complexity statement.

10.3 A precise conditional algebraic route

Reduced Khovanov rank one characterizes the unknot [6]. The same rank-one decision is sound over $\mathbb{F}_2$: rank one over $\mathbb{F}_2$ forces rational rank at most one by universal coefficients, while reduced knot homology is nonzero; the rational rank-one theorem then applies. Conversely the unknot has reduced rank one over $\mathbb{F}_2$. The baseline’s local-complex and cancellation approach follows the practical framework of Bar-Natan [2]; a general small-complex bound is a separate requirement.

To turn that endpoint into a quasi-polynomial algorithm, a compression scheme needs a closed set of operations and a size theorem, not merely small examples or an existence statement.

Theorem 10.2 (Conditional summary criterion). *Let $Q(n) = \exp((\log(n+2))^c)$ for a fixed constant c . Suppose a proposed exact recognition method has, on every n -crossing input, the following proved properties:*

1. *it constructs at most $Q(n)$ summaries or intermediate states;*
2. *each summary has at most $Q(n)$ bits;*
3. *constructing, composing, restricting, simplifying, comparing, and checking a summary uses at most $Q(n)^{O(1)}$ bit operations;*
4. *a final predicate, evaluable within the same bound, is equivalent to reduced Khovanov rank one or to another complete unknot criterion.*

Then the resulting exact recognizer is quasi-polynomial time.

Proof. Multiply the number of operations by their cost. A fixed power of $\exp((\log(n+2))^c)$ is still quasi-polynomial. Exact invariants and the terminal equivalence imply correctness. $\square$

The theorem is intentionally a contract rather than a breakthrough: its hypotheses identify the work still missing. The Potts representations satisfy analogous storage and operation bounds at bounded width, but their terminal predicate is incomplete. The residue-polynomial theorem shortens a local certificate, but does not prove the global summary count or size. The earlier scalar splitter finds some endomorphisms but does not guarantee a useful decomposition on every input.

For an explicit ring-valued implementation, the coefficient space alone has p^b basis monomials. A block with multiplicity r can have r^2 entries. Bounding an inverse polynomial’s degree by pr leaves both p^b and the total number of blocks to be controlled. Using a circuit or shared DAG can reduce their representation, but then composition, equality and zero detection for that representation also require proved bounds. Constant-size formulas whose expansion is exponential do not by themselves satisfy the criterion.

10.4 The geometric route and the current literature

Lackenby’s 2021 slides announce a deterministic recognition algorithm with bound $2^{O((\log n)^3)} = n^{O((\log n)^2)}$ [7]. That is quasi-polynomial, but it is not the stronger $n^{O(\log n)}$ target sometimes used in project notes. The July 2026 hierarchy preprint supplies a new recognition algorithm and certification framework; its Section 9 separates iteration counting from the cost of operations and bounds iterations by $L(G+1)^L$, where L is maximum hierarchy length and G maximum surface pattern-complexity [8]. We use G here to distinguish it from the component frontier g .

Published compressed-curve algorithms compute intersection numbers with cost polynomial in triangulation size and logarithmic normal weights; the isotopy result additionally assumes essential

normal one-manifolds [9]. These provide concrete components for a geometric implementation. Neither a fast terminal-pattern check nor an iteration count by itself supplies a complete bound on construction, cutting, simplification, representation growth and restarts.

For example, an independently proved bound $L = O(\log n)$ and $G = n^{O(1)}$ would make $L(G + 1)^L = n^{O(\log n)}$; with $L = O((\log n)^2)$ it would give $n^{O((\log n)^2)}$. These are conditional substitutions, not bounds established for the hierarchy inputs generated by this package. Likewise, a bit cost polynomial in an expanded normal weight is inadequate when that weight is exponential in its encoded length.

The strongest credible program therefore has two interacting branches: exact algebraic obstructions and compressed homological decisions for efficient practical rejection, and fully costed compressed geometric operations for a possible general recognition guarantee. They should share validated inputs, deterministic evidence records and resource-safe orchestration, while retaining separate mathematical obligations.

11 Further research questions and concrete next experiments

The following program distinguishes short implementation projects, structural mathematical questions, and obligations needed for a full complexity theorem. A successful next step should produce either a proof with explicit hypotheses, an independently replayable improvement, or an informative counterexample.

11.1 Immediate work on scalar transfers

Research question 11.1 (Optimize the actual component cost). Can crossing orders and checkerboard colors be chosen to minimize merge work, rather than only the global cut-edge or active-face maximum?

The current shading heuristic minimizes $(\max f_i, \sum f_i)$, while Theorem 5.2 depends on g and actual work also depends on alignment counts. A useful objective combines the sum of component orbit counts, the products at merge steps, and estimated integer lengths. Compute these structural features without running the full tensor contraction, then compare greedy local changes on a fixed corpus. The benchmark should include construction time and preserve orders on which the existing heuristic wins. A worst-case approximation guarantee for this objective would be more valuable than a favorable average alone.

Research question 11.2 (Switch representations at justified points). Can a certified runtime policy choose among matching, one-tensor Potts, and component Potts without paying their full combined costs?

The present measurements disprove a universal dominance claim. A first policy can select an algorithm from cheap structural profiles and give each attempt a fixed work budget. A stronger policy would convert a partially computed representation when its cost grows. Such a conversion needs an explicit linear map and an exact normalization account. Training a heuristic on these 29 examples and reporting its accuracy on the same examples would not answer the question; use held-out diagrams and relabeling controls.

Research question 11.3 (Factor beyond processed connected components). When can an exact active tensor be split along articulation vertices, small separators, or algebraic low-rank structure?

Graph-component independence is sufficient but not necessary for a compact tensor. At a one-vertex separator, condition on the separator spin and exploit the remaining color stabilizer. At larger separators, equality patterns induce different stabilizer groups and orbit multiplicities. The first milestone is a complete merge-and-forget calculus for separators of size one and two, checked against labeled enumeration. Algebraic rank factorization must preserve signed exact coefficients and must charge for rank discovery. A large output rank supplies a natural counterexample to an overly broad compression conjecture.

Research question 11.4 (Use contraction trees without assuming disk frontiers). Can a tree-based exact Potts contraction improve path-order bottlenecks in the repository while retaining the symmetry quotient?

Existing tensor and graph-width literature provides the starting point [11, 12]. The implementation question is how to combine separator assignments, spin orbits and exact coefficient growth in a reproducible cost model. A contraction tree should be stored as part of the computation input and validated. Compare total contractions, maximum tensor size, integer lengths and order-search cost. A tree decomposition is useful only if its conversion and join operations are also bounded.

11.2 Detection strength and exact specializations

Research question 11.5 (Characterize the kernel of the six-color evaluation). Which nontrivial knots, if any, have $V_K(2 + \sqrt{3}) = 1$?

This is a structural question about a specific evaluation. Start with families having exact trace or skein recurrences: closed braids of fixed index, twist constructions, satellites with controlled pattern, and connected sums. Search for polynomial divisibility by $t^2 - 4t + 1$ using exact Laurent polynomials. A found collision is valuable even if another filter already detects the knot. Failure to find one in a knot table gives no completeness theorem.

Research question 11.6 (Design a small complementary menu). Can a bounded or slowly growing set of algebraic points eliminate broad classes of collisions while keeping total work small?

The $q = 5$ failure and $q = 6$ repair show why algebraic complementarity matters. Measure additional detections per unit of exact work, with first-passing and never-detected cases recorded. Distinguish changing q from changing a prime at fixed q . Proposition 10.1 gives an unconditional growing menu for detecting whether the full Jones polynomial is 1; improving its number of points on a proved class is a concrete target. A universal fixed menu would require a new theorem about the possible Jones polynomials of knots.

Research question 11.7 (Arithmetic circuits or reconstruction). Can modular evaluation and certified reconstruction outperform integer pairs for very long, narrow diagrams?

The exact bit bound supplies an a priori reconstruction threshold. One could evaluate over enough compatible primes, reconstruct each coordinate, and verify with an additional exact identity. It is essential to count prime selection, conversions, and total bit work. Pair arithmetic has unusually cheap local multipliers, so a modular strategy may lose on moderate inputs. Large-parameter tests should use the same diagram order and report integer sizes, not merely crossing count.

Research question 11.8 (Shorter certificates than full replay). Can a nontrivial exact Potts value be certified with a verifier materially faster than recomputing the transfer?

The current evidence stores inputs and exact outputs, so replay has the same complexity class as evaluation. A contraction transcript can expose local sums and products, but may be as large as the computation. Explore checkable arithmetic circuits, deterministic residual identities, or interactive-verification models with their assumptions stated. A soundness claim must account for arbitrary forged intermediate tables, not just altered final values.

11.3 Ring-valued compression for the complete fallback

Research question 11.9 (Find useful ring-valued chain endomorphisms). Can the scalar candidate search from the preceding continuation be extended to produce small nontrivial same-matching ring-valued endomorphisms?

Theorem 9.4 assumes the operator is supplied. A concrete experiment is to add low dot-degree unknowns to the scalar intertwining equations, solve them exactly, and test whether the lifted Fitting summands reduce work on actual scanner blocks. Record the search dimension, solution support, block sizes before and after splitting, and the cost of checking all categorical cross terms. A search that routinely costs more than the saved elimination is not a performance improvement.

Research question 11.10 (Exploit low annihilator degree). When is the residue characteristic polynomial substantially larger than a usable annihilator, and can a shorter one be found cheaply?

The universal certificate has degree pr , and the radical cutoff is sharp in the full class. Particular matrices may have much smaller residue minimal polynomials or additional relations. A residue minimal polynomial alone need not lift with exponent p by the proof of Theorem 9.1; that proof uses characteristic-polynomial coefficients over the large ring. The first task is to identify a valid lifting theorem or an explicit counterexample before replacing the characteristic polynomial in code. Compare any successful refinement with doubled geometric-series inversion.

Research question 11.11 (Keep Fitting projectors succinct). Can $e(A)$ and its basis changes be manipulated without expanding their coefficient supports or multiplicity spaces?

A degree- $O(r)$ polynomial is short, but evaluating it can create large entries. Candidate representations include sparse monomial maps, shared composition DAGs, and block recurrences. Each needs exact addition, composition, zero testing, and application to differentials. The target is a closure theorem bounding all these operations on a specified class. Measure both the representation size and the cost of operations performed on it; a small serialized expression can still hide expensive evaluation.

Research question 11.12 (Decision information without full homology). Which exact summaries retain the rank-one question under all tangle extensions while discarding irrelevant homology?

Previous saturated decisions and homological windows suggest useful restricted summaries, but a complete invariant of future continuations can be much larger than a scalar Euler characteristic. Formulate a congruence on partial complexes that is respected by every allowed crossing extension. Prove that the terminal decision descends to the quotient. Then either bound the number of quotient states or exhibit a family requiring many. This is a direct route toward the missing hypotheses of Theorem 10.2.

Research question 11.13 (Learn from the explicit three-braid barrier). Which features permit compressed three-braid algorithms, and which survive at four or more strands?

The 2026 results provide both a warning about explicit output size and a positive compressed model [5]. A useful next project is to encode a proven fixed-strand family with exact multiplicities and verify the closed recurrences against the baseline scanner. The research target is an extension theorem with a precise parameter, not an empirical claim that all fixed-width complexes remain small.

11.4 A fully costed geometric branch

Research question 11.14 (Implement compressed normal-curve operations). Can the published intersection and isotopy algorithms be integrated with ProveIt’s boundary-pattern representation?

The initial deliverable should handle explicit triangulated surfaces, binary normal coordinates, geometric intersection numbers and the essential curve precondition for isotopy [9]. Tests must include exponentially large weights described by short binary inputs. Enumerating all normal arcs would destroy the intended complexity. Every conversion between patterns, triangulations and normal coordinates must be part of the time and size account.

Research question 11.15 (Control hierarchy restarts and encodings). What invariant gives a useful simultaneous bound on hierarchy length, pattern complexity, representation size and restart cost?

The expression $L(G + 1)^L$ motivates a quantitative target but is not itself a complete running-time theorem. Instrument an implementation of one well-defined hierarchy step and prove that its compressed output meets a stated bound. Then analyze how cutting and simplification interact with that bound under restarts. A fast terminal checker is already a useful component; the next milestone is a fully costed nonterminal transition, followed by a global potential argument.

Research question 11.16 (Connect the algebraic and geometric branches). Can an exact algebraic obstruction or partial homological summary guide geometric choices without becoming an unproved pruning rule?

A safe interaction uses algebraic information to prioritize work while retaining every branch required by the geometric correctness proof. A stronger pruning rule needs its own implication theorem. The first practical milestone is shared diagram and certificate formats with explicit preprocessing equivalences. The longer-term milestone is a combined algorithm whose worst-case bound can be proved independently of its heuristic ordering choices.

11.5 Evaluation discipline and recommended sequence

Before a broad default change, extend the corpus beyond small named fixtures and short random braids. Include pairs of diagrams of the same knot with different embeddings and orders, nontrivial Alexander-trivial knots, difficult unknot diagrams, large connected sums, and families chosen to stress each representation. Record cases solved before the Jones stage separately. Measure failed or budget-limited attempts as costs, not as missing data.

The recommended sequence is: first improve the component-aware order objective and confirm it on held-out inputs; next extract actual same-matching blocks to test the residue certificates; then implement one compressed geometric operation with a complete bit-complexity contract. In parallel, maintain the exact-specialization collision search and the independent oracles. This sequence produces useful verified increments while keeping the eventual quasi-polynomial theorem tied to explicit, testable obligations.

12 Reproducibility, integration, and the final research position

12.1 Contents of the archive

Location	Contents
<code>paper/</code>	This PDF, the main LaTeX source, all section sources, BibTeX references, generated tables, vector figures and preview PNGs.
<code>fast/</code>	The modified implementation, all existing fixtures and tests, four Potts modules, integration tests and the new backend guide.
<code>reference/</code>	The selected pinned repository snapshot, including the baseline implementation and research notes used for the audit.
<code>integration/</code>	The patch against the pinned tree, a list of changed files, source provenance and Git blob identifiers.
<code>research/</code>	Independent smoothing-cube, finite-menu and truncated-ring oracles; proof-audit notes; certificate replayer; saved verification records and baseline/modified test logs.
<code>benchmarks/</code>	Kernel and normal-pipeline scripts, all raw samples, CSV summaries, source-hash audits, measured module versions and figure-generation code.
<code>certificates/</code>	Example exact $q = 6$ obstruction certificates, the W_5 collision record, and saved replay results.
<code>tools/</code>	Manifest and patch verification, certificate regeneration, and a reproduction driver that writes to a fresh run directory.
<code>SHA256SUMS</code>	SHA-256 digest list for distributed files.

The source snapshot contains 98 selected text files whose Git blob hashes were checked at retrieval. It is not represented as a complete clone of the entire ProveIt repository. The integration patch affects only the `Topology/UnknotRecognition/fast/` subtree. It adds the new modules and tests, changes the Jones dispatch and CLI budget handling, and documents the options. It leaves the categorical fallback implementations intact.

12.2 Reproducing the mathematical and software checks

From the root of the extracted archive:

```
python tools/verify_manifest.py
python tools/verify_integration.py
python tools/reproduce.py
```

The manifest checker verifies both archive digests and pinned Git object hashes. The integration checker applies the patch to a disposable copy and compares the resulting source bytes with the delivered implementation. The reproduction driver runs the integrated unit suite, independent oracles, replay self-tests and supplied certificates in a new directory. It preserves the distributed result records and writes commands, durations, exit codes and logs for the new run.

To repeat the timings, the large-integer check and the PDF build as well:

```
python tools/reproduce.py --all
```

Timing outputs should not be expected to match the distributed nanosecond samples. Exact values, deterministic work counters, test results and certificate verdicts should match. Differences in Python version, machine load and platform are recorded by the scripts.

Core algorithms and validation use the Python standard library. The implementation requires Python 3.10 or later; the recorded environment used 3.12.14. Regenerating the figures requires Matplotlib. The paper uses pdf $\text{\LaTeX}$, Bib $\text{\LaTeX}$, latexmk and the packages listed in its preamble. The complete source is included so that layout, notation, references and integration conventions can be revised in the repository.

The independent oracles are intentionally small-instance tools. The smoothing cube is exponential in crossing number. The truncated-ring oracle uses explicit monomials and small matrix characteristic polynomials. They test the optimized algorithms through a different representation; they are not proposed as scalable alternatives.

12.3 Integration procedure

At a checkout of the pinned ProveIt revision, run:

```
git apply --check /path/to/package/integration/changes.patch
git apply /path/to/package/integration/changes.patch
```

The patch has been checked by applying it to the included reference copy and comparing all resulting implementation files. A later repository revision may require an ordinary merge. The article and its associated research artifacts can be placed in a new report directory using the project’s preferred numbering.

The package retains the source’s MIT No Attribution license and supplies the continuation code and documentation under the same license. Scholarly sources are cited through their primary papers and author or publisher pages; the archive does not redistribute copies of those papers. This is a mathematical research draft with reproducible experiments, not a proof-assistant formalization.

12.4 What the continuation establishes

There is a concrete, verified scalar improvement: fixed-color equality patterns and component independence give a smaller, precisely bounded representation for a Jones obstruction. Exact quadratic arithmetic makes its bit cost and equality semantics explicit. The five-color weaving family shows why exact arithmetic alone is insufficient, while the six-color calculation gives a proved repair of that family. The growing menu gives a separate quasi-polynomial Jones-triviality decision on logarithmic-width orders.

There is also a precise local algebraic advance: residue characteristic polynomials give bounded-degree annihilators, sharp radical cutoffs and constructive Fitting certificates for supplied commutative endomorphism blocks. These results extend the available tools without assuming that the entire categorical computation remains small.

The next general recognition theorem must still connect an exact complete criterion to a globally bounded representation and a fully costed search. The archive is intended to make the present increments usable while keeping that larger obligation visible and mathematically testable.

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